RAGHAVENDER REDDY BADUGULA

Mechanical Engineer | CFD & FEA Simulation | Materials R&D | MSc Cranfield University

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PROFESSIONAL SUMMARY

MSc Advanced Mechanical Engineering graduate from Cranfield University with practical experience in CFD, FEA, mechanical design, and materials research. Experienced in ANSYS Fluent, ICEM CFD, Abaqus, SolidWorks, and MATLAB, with experimental and computational research covering high-temperature molten salt corrosion of WAAM stainless steel, biomimetic airfoil optimization, and offshore wind monopile buckling analysis. Qualified in GATE 2022 (Smart Manufacturing) with UK operational leadership and technical project coordination experience. Seeking Mechanical Engineering, Simulation (CFD/FEA), Materials R&D, and Project Engineering roles in the UK.

CORE ENGINEERING & TECHNICAL SKILLS

- **CFD, FEA & Numerical Simulation:** ANSYS Fluent (2D/3D Viscous Flow, k-omega SST Turbulence Modeling), ANSYS ICEM CFD (Multi-Block Structured O/C-Grid Meshing, y+ Boundary Layer Sizing, 4-Stage Mesh Independence), Abaqus FEA (3D Shell Partitioning, Structured Quad Meshing, Linear Perturbation Eigenvalue Buckling), MATLAB (Iterative Blade Element Momentum [BEM] Solver, Airy Linear and Non-Linear Offshore Wave Kinematics), XFOIL.
- **Materials, Corrosion & Laboratory R&D:** Wire Arc Additive Manufacturing (WAAM SS316L), High-Temperature Molten Salt Corrosion Testing (600°C Ternary NaNO3/KNO3/KCl), SEM-EDS Cross-Sectional Elemental Mapping and Point-ID Spectroscopy, 3D Optical Surface Profilometry (30-Position Stitched ZVI Interferometry), Metallographic Sectioning/Mounting/Polishing, Precision Gravimetry (10^-5 g), Marine Coating Hypothesis Testing, COSHH Risk Assessments.
- **Mechanical Design & Project Engineering:** SolidWorks (3D CAD Assemblies, Mechanism Kinematics), AutoCAD, Rapid 3D Printing (Bambu Studio PLA Slicing & Prototyping), Project Charter Governance, Quantitative PM Risk Assessment Matrices (Probability-Impact Scoring), Multi-Phase Gantt Scheduling, UK/ISO Standards Compliance (BS EN 12464-1, ISO 3382-3, BS EN 1729-1, BS 8300, DNV-GL-ST-0126).
- **Software & Automation Tools:** Python, MATLAB Scripting, Git/GitHub Actions CI/CD, Playwright Automated Testing, Technical Documentation and Data Visualization.

EDUCATION & QUALIFICATIONS

MSc in Advanced Mechanical Engineering | Cranfield University, UK — Oct 2024 to Sep 2025

- Top Module Achievements: Fluid Mechanics and Offshore Loading (74%), Component Design and Prototyping (73%), Structural Integrity (69%).
- Core Research & Modules: High-Temperature Molten Salt Corrosion of WAAM SS316L (MSc Thesis), CFD for Renewable Energy (ANSYS ICEM CFD & Fluent), Engineering Stress Analysis (Abaqus FEA), Assessing Risk & Failure, Industrial Group Project with Nissan, Engineering Design & Project Management.

B.Tech in Mechanical Engineering (Mechatronics) | MGIT, Hyderabad, India — Aug 2016 to Nov 2020

- GATE 2022 Qualified: Smart Manufacturing and Advanced Production Systems. Focus: Mechatronics, CAD/CAM, Robotics Kinematics, and Thermo-Fluid Systems.

FEATURED ENGINEERING, SIMULATION & R&D PROJECTS (CRANFIELD UNIVERSITY & MGIT)

Molten Salt Corrosion of WAAM SS316L for CSP Thermal Energy Storage (MSc Thesis) | Cranfield University — 2024 to 2025

- Subjected Wire Arc Additively Manufactured (WAAM) SS316L specimens to ternary chloride molten salts (10NaNO3-78KNO3-12KCl) at 600°C for 250 h and 500 h in a horizontal tube furnace under nitrogen purge, comparing open vs. covered-lid crucibles to simulate vapour-retentive Concentrated Solar Power (CSP) tank zones.
- Prepared metallographic cross-sections through abrasive sectioning, cold-resin mounting, sequential SiC grinding, diamond polishing, and ultrasonic cleaning with acetone and isopropanol (IPA).
- Conducted 30-position stitched 3D optical dimensional metrology (.zvi surface profilometry and cumulative probability radius-change distributions), precision mass change metrology (10^-5 g accuracy), and cross-sectional SEM-EDS elemental mapping and Point-ID spectroscopy.
- Demonstrated that covered-lid vapour retention accelerates porous scale formation, Na/K/Cl/O ingress, and localized pit initiation at 250 h compared to compact Cr/Fe-rich oxides in open crucibles, and authored full COSHH risk assessments and high-temperature safety protocols.

ICEM CFD Aerodynamic Simulation & Biomimetic Airfoil Optimization for Urban VAWTs | Cranfield University — ANSYS ICEM & Fluent

- Investigated the aerodynamic performance of a modified NACA 0015 airfoil featuring a 40%-chord upper-surface bump and a stingray-inspired trailing edge for urban Vertical Axis Wind Turbines (VAWTs) at 4 m/s and 12 m/s.
- Developed a multi-block O-grid structured hexahedral mesh in a C-shaped domain (-20c to +40c) using ANSYS ICEM CFD (first-layer spacing 3.3x10^-6 m for target y+, 2x2x2 determinant quality > 0.837) and completed a 4-stage mesh independence study from 21,274 to 104,274 nodes (<1% variation).
- Performed 2D viscous flow simulations using the k-omega SST turbulence model in ANSYS Fluent, demonstrating delayed upper-surface boundary layer separation and increasing lift coefficient (CL) from 0.80 to 0.95 at 10° angle of attack.

FEA Buckling & Stress Analysis of Offshore Wind Turbine Monopile Foundations | Cranfield University — Abaqus FEA

- Evaluated structural instability and critical buckling loads of a 40 m offshore wind turbine hollow steel monopile (outer radius R = 3.75 m, wall thickness t = 0.08 m) in Abaqus using 3D shell elements partitioned across atmospheric, seawater, and seabed zones with fixed encastre base boundary conditions.
- Constructed a 23,200-element structured rectangular quad mesh (0.2 global seed) to eliminate tetrahedral discretization noise and executed Linear Perturbation (Eigenvalue Buckling) analysis under combined eccentric rotor/nacelle weight, wave pressure (1.5 kPa), and seabed pressure (20.3 kPa).
- Extracted the first 5 critical buckling modes (eigenvalues lambda_1 = 0.3349 to lambda_5 = 0.5917), characterizing progressive multi-wave shell instabilities and fixed-base stress concentrations against DNV offshore structural standards.

Fluid Mechanics & Offshore Loading: HAWT BEM Rotor & Airy Wave Kinematics | Cranfield University (74%) MATLAB & XFOIL

- Wind Turbine Rotor Aerodynamics (Part A): Generated NACA 0030 airfoil lift/drag polars in XFOIL and wrote an iterative Blade Element Momentum (BEM) convergence solver in MATLAB for a 3-bladed Horizontal Axis Wind Turbine (R = 7 m), computing radial induction factors, total thrust (1,804.6 N), torque (1,693.8 Nm), power (10.16 kW), and power coefficient (Cp = 0.1872), alongside blade twist optimization.
- Offshore Hydrodynamic Wave Loading (Part B): Applied Linear (Airy) and Non-Linear wave theory (H = 0.6 m, T = 10 s, wavelength = 43.69 m, depth = 2 m) in MATLAB to simulate depth-dependent fluid velocity profiles and elliptical water particle trajectories (z = 0 to -1.9 m) for offshore monopile met masts.

Chemical Preparation of a Novel Semiconductor Solar Paint | Group Project with Nissan, Cranfield University 2024 to 2025

- Formulated a lead-free photovoltaic solar paint matrix by dispersing bismuth methyl iodide quantum dots (1.8 to 2.3 eV bandgap) within a TiO2 electron-transport matrix via controlled mechanical and magnetic stirring across 1x, 2x, and 3x precursor concentrations.
- Prepared galvanized automotive steel substrates and performed electrical conductivity and phase-stability testing over 48-hour cycles using calibrated multimeters and optical equipment, achieving a 5x conductivity enhancement in double-concentration samples while coordinating group report synthesis.

Ship Hull Marine Corrosion & Paint Layer Hypothesis Testing (Assessing Risk & Failure) | Cranfield University 2024 to 2025

- Designed and executed controlled marine corrosion hypothesis testing and risk/failure assessment comparing single, double, and triple ship hull coating thicknesses under simulated seawater exposure.
- Demonstrated through empirical statistical analysis that double-layer coating application seals microscopic pinholes to significantly optimize anti-corrosion protection efficiency over single-layer coating, while avoiding the curing stress and diminishing returns of triple-layer systems.

Smart & Sustainable Shower System: Component Design & 3D-Printed Prototype | Cranfield University (73%) SolidWorks & 3D Print

- Modeled in SolidWorks 3D CAD and 3D-printed (Bambu Studio PLA) a closed-loop shower cold-water recirculation prototype integrating temperature sensors, solenoid directional valves, an under-tub storage reservoir, and a return pump, targeting a 50% to 70% reduction in pre-heating water wastage.

Engineering Design & Project Management: Stafford Cripps SC3 Smart Classroom Modernization | Cranfield University Project Lead

- Authored the Project Charter, Double Diamond design justification, circular-economy furniture reuse plan, UK/ISO compliance specification (BS EN 12464-1, ISO 3382-3, BS EN 1729-1, BS 8300), 15-item quantitative PM Risk Assessment Matrix (reducing PI scores from 25 to <= 9), and 11-week Gantt schedule.

Multi-Terrain Quadruped Robot & Thermoacoustic Engine Simulation | Undergraduate Projects, MGIT 2016 to 2020

- Designed 3D quadruped mechanical leg assemblies and chassis structures in SolidWorks to evaluate walking gait kinematics and stability dynamics, and conducted thermal-acoustic wave simulation analysis for thermoacoustic energy conversion systems.

WORK & OPERATIONAL EXPERIENCE

Operational Specialist (Team Lead Progression) | Smashburger, Milton Keynes, UK Nov 2024 to Present

- Selected for Team Lead progression based on frontline operational performance in a high-volume UK hospitality environment.
- Coordinating shift workflows, quality control, food safety auditing, and inventory management across daily store operations.

Assistant Project Manager (Less than 1 Year) | Enterprise Assessment & Localization Operations 2023 to Sep 2024 (<1 Year)

- Coordinated end-to-end multilingual translation, Subject Matter Expert (SME) validation, and data entry workflows for confidential national-scale competitive examination and assessment projects.
- Managed cross-functional teams of freelance translators and typists across regional languages, ensuring 100% SLA deadline adherence and strict security compliance.

Freelance Technical Translator & Localization Specialist | Media & Corporate Clients 2018 to Sep 2024

- Delivered high-accuracy English-to-Telugu technical translation, subtitle localization for major streaming releases (Netflix and Amazon Prime), and remote interpretation for UK corporate and legal clients.

Industrial Engineering Internships | RINL Vizag Steel Plant & Maruti Suzuki (Varun Motors), India 2018 to 2020

- Completed industrial manufacturing and maintenance training at Rashtriya Ispat Nigam Limited (RINL Vizag Steel Plant: Coke Plant and Medium Merchant & Structural Mill I-beam rolling) and Maruti Suzuki automotive service operations.

INDEPENDENT SOFTWARE & AUTOMATION PROJECTS

Polyglot Engine (github.com/Raghavender1234/polyglot-engine), IndiCAT (indicat.co.in) & OpenLMS (github.com/Raghavender1234/pr-dia) Systems Logic & Product Architecture

- Polyglot Engine CLI: Designed the 6-layer system architecture and closed-loop repair logic for an open-source multilingual UI layout scanner (Playwright DOM geometry, HarfBuzz text shaping, Unicode BiDi) built via AI-assisted software workflows.
- IndiCAT Workstation & OpenLMS: Conceived and shipped a multi-format Windows CAT and localization app on the Microsoft Store (offline IndicTrans2/NLLB, Whisper dubbing, Translator CRM) and an open-source multi-tenant engineering assessment LMS.